(Minor, 2022)(LE ONN) Safety

Driver not paying enough attention while driving

Solution

Create application to make sure the driver has best performance

(This feature helps drivers recognize when their attention to the road is becoming diminished, enhancing their awareness.)

https://www.hondainfocenter.com/2023/CR-V/Feature-Guide/Interior-Features/Driver-Attention-Monitor/

(With the unstoppable rise of EV's, the AI installed in the EV can drive itself safely along the road, avoiding and overtaking other incoming vehicles. A good example would be Tesla's AI technology called Autopilot. Autopilot)

Problem

Many accidents are caused by drivers not paying attention while driving. For example, many drivers suffer from driver fatigue when travelling long distances. Driver fatigue is mainly caused by a lack of sleep, causing the loss of awareness on the road. For example, on December 31st 2022 a truck driver in Tampa, Florida dozed off as recorded by his truck's dash-cam (Minor, 2022). He collided with a car parked on the side of a highway where two people were replacing a tyre. Thankfully, no deaths occurred.

Besides, some irresponsible drivers tend to text/call while driving, distracting themselves from paying attention on the road. In fact, this is such a serious issue in the US that 1 in 4 car accidents there are caused by texting and driving. When we call or text while driving, this causes a cognitive distraction where our minds are focused on the conversation. We may not have sufficient time to react when facing a oncoming hazard on the road. Hence, we should not pick up calls unless we are having an emergency.

While self driving cars are seen as mainstream now, the truth is that most automated vehicles are only partially automated. The development of self driving cars are only in its primary stages and is incapable of fully controlling itself. Researchers in the University of Toronto has discovered a phenomenon where many drivers seem to be over reliant on the

automation system because of the in-vehicle display ‘TORAC’ (takeover request and automation capability) (P., 2020) .Whenever the AV's automation system fails, the driver will be asked to retake control of the vehicle. This creates the illusion where the driver thinks he plays an active role in driving, although the automation system does most of the part. For example, Tesla Autopilot has been involved in 736 car accidents in the US since 2019 (Kothari, 2023). Since May 22, there have been 17 more fatal crashes where 11 of them caused deaths. These statistics show that self driving EV’s can cause drivers to be layback and trust the autopilot system too much.

Solution

Such accidents can be prevented with the help of technology, where companies can develop an application to make sure drivers are in the best conditions when driving. This feature can help drivers recognise when their attention on the road is being diminished, enhancing their senses to lower the chances of getting involved in an accident. For example, car company Honda has developed a safety feature called the Driver Attention Monitor (quote, Honda Info Centre, n.d.). This feature uses angle sensor to measure the amount of wheel corrections made by the driver to maintain good lane position. If the system detects too many corrections made by the driver, a beep sound will be made and the steering wheel will vibrate. This can help reduce accidents from occurring.

Besides, EV companies can improve the device-car integration to ensure drivers stay focused while driving. Google has developed a mobile app called Android Auto, where it allows you to connect your phone with your car's display system when plugged in with a USB cord. Users are allowed to use maps, make calls or control media playback. The main idea is to display large icons, buttons, and menus that are easy to select and also use voice as a primary input method (Conway, 2023). Drivers can install apps that will block incoming calls or messages when it detects the car is in motion. In case of an emergency, drivers can choose to answer the phone via Bluetooth technology which allows hand free calling. In this way, drivers can focus on the road with both hands on the steering wheel, minimizing the risk of getting in a car crash.

Over reliance on AV's autopilot system seems to be an emerging problem. However, AV producers can use facial recognition technology to ensure the driver is focusing on the road while the car's autopilot system is on. American company General Motors created a driver assistance technology 'Supercruise' where it uses a camera to monitor whether the driver's head and eye position is focused on the road (SUPER CRUISE™, n.d.). If not, the light on the steering wheel will start flashing red and emit a beeping sound, followed by a voice prompt to tell you to take control of the wheel after a while. If attention is still not given, the car will automatically slow down and park itself on the side of the road. An emergency call centre will also be contacted to ensure you are safe. Besides, Tesla's Autosteer system

requires drivers to always put both hands on the steering wheel (Autopilot and Full Self-Driving Capability, n.d.). If not enough force is applied, you will automatically be reminded through audio and visually to ensure control of the steering wheel.

In conclusion, the advances of technology creates new opportunities but also creates problems. However, with the right amount of creativity we can solve all the problems faced.

References

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With the unstoppable rise of EV's, the AI installed in the EV can drive itself safely along the road, avoiding and overtaking other incoming vehicles. A good example would be Tesla's AI technology called Autopilot (S., 2023). Autopilot allows the car to drive itself, avoid incoming vehicles and stay on its lane. While self driving cars is a admirable innovation, it has also caused many people to be over reliant on the car's autopilot system. Many drivers tend to not pay attention on the road and as a result get involved in and accident whenever an error occurs in the car's system. For example, Tesla Autopilot has been involved in 736 car accidents in the US since 2019 (Kothari, 2023). Since May 22, there have been 17 more fatal crashes where 11 of them caused deaths. Such statistics show that self driving EV's can cause drivers to be layback and trust the autopilot system too much. Tesla's Autopilot system also has improvements to be made.